AFM®

## Product ID:

| | |
|---|---|
| **Name:** | Dryden Aqua AFM® - Activated Filtration Media |
| **Usage:** | Replacing sand in all media filtration applications |
| **Material:** | Green recycled glass, uniquely shaped particles with activated surface |
| **Unique Features:** | Bio-resistant, self-sterilizing, sustainable, activated filtration media |

## Main Benefits:

- Over 30% lower running costs vs. sand
- Over 50% lower chlorine consumption and lower disinfection by-products vs. sand
- Filtration down to several microns at 100% efficiency without filtration aids
- More than 4 times longer lifecycle than sand
- Specifically engineered product, targeted for an efficient removal of priority metal oxides and other target constitutes
- Guaranteed prevention of channeling & biological fouling on AFM® surface

nqa. ISO 9001 Registered

UKAS QUALITY MANAGEMENT 015

TESTED AND CERTIFIED UNDER INDUSTRY STANDARDS WaterQuality ®

NSF/ANSI 61

## About Dryden Aqua AFM®:

AFM® replaces sand in all filtration applications. It is manufactured from a specific glass type and processed to obtain the optimum particle size and shape, and then activated to increase the surface area by 300 times over crushed glass or sand.

The high surface area is negatively charged (zeta potential) to electro-statically attract organics and small particles. It also has permanent metal oxide catalysts, creating a high redox potential, to make AFM® self-sterilizing.

DRYDEN AQUA SUSTAINABLE WATER QUALITY

## How to use AFM®

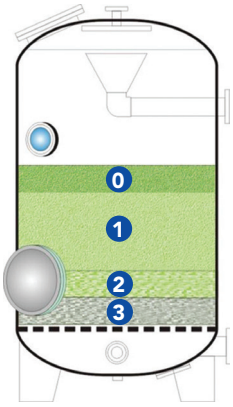
- Simply replace the sand in your media filters with AFM®
- AFM® has a 15 % lower density than sand by weight:

  E.g. if your filter takes 200 kg of sand it will only require: 200 x 0.85 = 170 kg of AFM®.

### Recommended operational parameters

| Parameter | Value |
|---|---|
| Filtration velocity | 1-15 $m^3/hr/m^2$ |
| Back Wash velocity | 35-55 $m^3/hr/m^2$ |
| Max. operating differential pressure | > 0.4 bar |
| Back Wash Duration | for 5-10 mins |
| Water pH limits | 3-9 |
| Water temperature limits | 1-100° C |

### Recommended AFM® Grades and Arrangement from top to bottom (on a bottom plate), in Pressure Filters




| Grade, Size (mm) | | Commercial Purification | High Purification | Ultra Purification |
|---|---|---|---|---|
| Grade 0 | 0.25-0.50 | n/a | 20 % | 60 % |
| Grade 1 | 0.4-1.0 | 70 % | 50 % | 20 % |
| Grade 2 | 1-2 | 15 % | 15 % | 10 % |
| Grade 3 | 2-6 | 15 % | 15 % | 10 % |

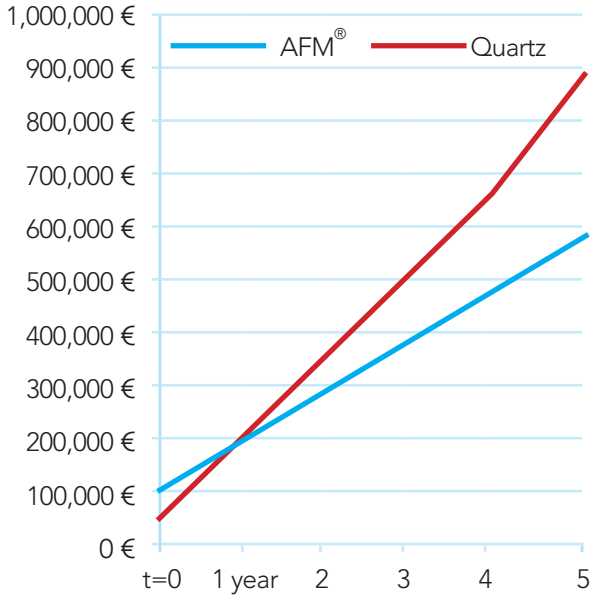
### Before first use of AFM®:

Back Wash the AFM® for 10-15 minutes, 3 consecutive times, with potable water and then rinse it for 1 hour.

### Recommended applications for Dryden Aqua AFM®

| Application Type | AFM® removes, for example: |
|---|---|
| Drinking water | Iron, Manganese, Arsenate and some priority chemicals |
| Municipal Wastewater | Phosphorous & Bacteria, BOD, COD & TOC |
| Industrial Process Water | Organic pollutants & oils, TSS, VSS & Particles < 1 micron |
| Industrial Wastewater | Colours and fine fibers, Residual mineral oils |

### AFM® is Cost-Effective: Lifecycle Cost Analysis

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